Quantum Fault Tolerance Meets Toom's Contours

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1 Shrinking

consider changing the name to something like hight lines / contour in order to hint on the similarity to topography map. In addition, the embedding inside $\mathbb{R}^d$ should appears before that definition.

Definition 1.1 (Potential Walls)**.** We define the potential walls to be a functions collection from the vertices of $\mathcal{G}_n^d$ to the reals as follow:

1. For any coordinate $i \in [d]$, we define $L_i \colon \mathcal{G}_n^d \to \mathbb{R}$ to act as $L_i(v) = v_i$.
2. For any subset of coordinates $I \subset [d]$ we define $L_I \colon \mathcal{G}_n^d \to \mathbb{R}$ to act as $L_I(v) = -\sum_{i \in I} v_i$.

For a subset of vertices $\Gamma \subset \mathcal{G}_n^d$, a vertex $v \in \Gamma$, and a potential wall $l \colon \mathcal{G}_n^d \to \mathbb{R}$, we say that v is a *local maximum* of l in Γ if for any neighbor u of v, such that $u \in \Gamma$, we have $l(v) \geq l(u)$. We say that v is a strong local maximum if the inequality is strict, namely $l(v) > l(u)$. When the subset Γ is clear from the context, we might omit it and briefly say that v is a local maximum of l.

What we want is that if a bit contains both v, gv then there will be a check that contains both v, gv, that is true if $d' > 1$, then it means that we can earse from the bit a vertex which it is not gv.

Claim 1.2. Let z be an assignment, and let ∂z be the set of vertices adjacent to unsatisfied checks induced by z. Let v be a vertex in ∂z, and let g be a coordinate in S such that v is a local maximum of L_g. Then $gv \notin \partial z$.

Proof. Suppose $gv \in \partial z$, then

$$L_g(gv) = (gv)_g = 1 + v_g = 1 + L_g(v)$$

In contradiction to v being a local maximum of L_g in ∂z. $\square$

Claim 1.3. Let z be an assignment, and let ∂z be the set of vertices adjacent to unsatisfied checks induced by z. Let v be a vertex in ∂z, and let g be a coordinate in S such that v is a local maximum of L_g. Then for any check associated with a $d'-1$ face rooted at v that contains gv, it is satisfied.

Proof. Assume, through contradiction, the existence of an unsatisfied check associated with a $d'-1$ positive face rooted at v, which contains gv. This implies $gv \in \partial z$, contradicting Claim 1.2. □

Definition 1.4 (Small Code at Vertex v)**.** For a vertex v, we define the X-*small code at vertex* v to be the binary assignments over the bits (d' faces) rooted at v that satisfy $\mathcal{C}_X$'s restrictions rooted at v ($d'-1$ faces). Similarly, we define the Z-*small code at vertex* v to be the binary assignments over the bits (d' faces) rooted at v that satisfy $\mathcal{C}_Z$'s restrictions rooted at v ($d'+1$ faces). We denote by $\mathcal{C}_v^-$ and $\mathcal{C}_v^+$ the X-small code at v and Z-small code at v.

Definition 1.5. Should rewrite it better. We define Stab_v^- and Stab_v^+ as the following linear codes:

$$\begin{aligned}\mathrm{Stab}_v^- := \big\{ z : \partial z = 0 \text{ and } \\ z \in \mathrm{Span}\left(\left\{\theta_{\mathbf{v},\mathbf{S'/g}}, \theta_{\mathbf{gv},\mathbf{S'/g}} \text{ for } S' \subset S, |S'| = d'+1, g \in S'\right\}\right)\big\}\end{aligned}$$

$$\begin{aligned}\mathrm{Stab}_v^+ := \big\{ z : \partial^* z = 0 \text{ and } \\ z \in \mathrm{Span}\left(\left\{\theta_{\mathbf{v},\mathbf{S'/g}}, \theta_{\mathbf{gv},\mathbf{S'/g}} \text{ for } S' \subset S, |S'| = d'+1, g \in S'\right\}\right)\big\}\end{aligned}$$

[COMMENT] Regarding the beneath, The number of restriction in ∂^+ case was counted in negligently. In addition, we should show that for the ∂^- case, restrictions of the form $\theta_{\mathbf{g_2 g_1 v},\mathbf{S}}$ are degenerate. In the bottom line it's look like:

$$\partial^+\theta_{\mathbf{g_2g_1v},\mathbf{S}} = \partial^+\theta_{\mathbf{g_1v},\mathbf{(S/g_3)\cup g_2}} + \partial^+\theta_{\mathbf{g_2v},\mathbf{(S/g_3)\cup g_1}}$$

Lets develop it, consider a restriction rooted at g_2g_1v, and let's look on it's boundary:

$$\partial^+\theta_{\mathbf{g_2g_1v},\mathbf{S'}} = \sum_{g\in S/S'} \theta_{\mathbf{g_2g_1v},\mathbf{S'\cup g}} + \sum_{g\in S/S'} \theta_{\mathbf{g^{-1}g_2g_1v},\mathbf{S'\cup g}}$$

Now, any term in the left closure is a bit beyond the scope of the bits in Stab_v^-, thus, we are zeroing them out. The bits, on the right closers are in the scope, only if $g \in \{g_1, g_2$. Thus, in total the check is equivalence to:

$$\begin{aligned}\partial^+\theta_{\mathbf{g_2g_1v},\mathbf{S'}} = \theta_{\mathbf{g_1v},\mathbf{S'\cup g_2}} \cdot \mathbf{1}_{g_2\notin S'} \\ + \theta_{\mathbf{g_2v},\mathbf{S'\cup g_1}} \cdot \mathbf{1}_{g_1\notin S'}\end{aligned}$$

$$\partial^+\theta_{\mathbf{g_1v},\mathbf{(S'/g_3)\cup g_2}} = \sum_{g\in S/S'} \theta_{\mathbf{g_1v},\mathbf{(S'/g_3)\cup g_2\cup g}} + \sum_{g\in S/S'} \theta_{\mathbf{g^{-1}g_1v},\mathbf{(S'/g_3)\cup g_2\cup g}}$$

$$\partial^+\theta_{\mathbf{g_2v},\mathbf{(S/g_3)\cup g_1}} = \sum_{g\in S/S'} \theta_{\mathbf{g_2v},\mathbf{(S'/g_3)\cup g_1\cup g}} + \sum_{g\in S/S'} \theta_{\mathbf{g^{-1}g_2v},\mathbf{(S'/g_3)\cup g_1\cup g}}$$

$$\partial^+\theta_{\mathbf{g_1v,(S'/g_3)\cup g_2}} = \left(\sum_{g\in S/S'} \theta_{\mathbf{g_1v,(S'/g_3)\cup g_2\cup g}}\right) + \theta_{\mathbf{v,(S'/g_3)\cup\{g_2,g_1\}}}$$

$$\partial^+\theta_{\mathbf{g_2v,(S/g_3)\cup g_1}} = \left(\sum_{g\in S/S'} \theta_{\mathbf{g_2v,(S'/g_3)\cup g_1\cup g}}\right) + \theta_{\mathbf{v,(S'/g_3)\cup\{g_1,g_2\}}}$$

Other way:

$$\begin{aligned}\partial^+ \sum_{g'\in S/S'} \theta_{\mathbf{g'v,S'}} &= \sum_{g'\neq g\in S/S'} \theta_{\mathbf{g'v,S'\cup g}} + \sum_{g\in S/S'} \theta_{\mathbf{v,S'\cup g}} \\ &= \sum_{g'\neq g\in S/S'} \theta_{\mathbf{g'v,S\cup g}} + \partial^+\theta_{\mathbf{v,S'}}\end{aligned}$$

Where we use the fact that if $g = g'$ then the bit is beyond the scope. In addition:

$$\begin{aligned}\partial^+ \sum_{g'<g\in S/S'} \theta_{\mathbf{g'gv,S'}} &= \sum_{g'<g\in S/S'} \theta_{\mathbf{g'v,S'\cup g}} + \theta_{\mathbf{gv,S'\cup g'}} \\ &= \sum_{g'\neq g\in S/S'} \theta_{\mathbf{g'v,S'\cup g}} \\ &= \partial^+ \sum_{g'\in S/S'} \theta_{\mathbf{g'v,S'}} + \partial^+\theta_{\mathbf{v,S'}}\end{aligned}$$

Thus, we find more $\binom{d}{d'-1}$ restriction over the checks. [We need to show that those restrictions are independent]. In addition:

$$\partial^+ \sum_{S'} \sum_{g'<g\in S/S'} \theta_{\mathbf{g'gv,S'}} = 0$$

$$\partial^+\theta_{\mathbf{g_2g_1v,S}} = \partial^+\theta_{\mathbf{g_1v,(S/g_3)\cup g_2}} + \partial^+\theta_{\mathbf{g_2v,(S/g_3)\cup g_1}}$$

I don't succes to advance the idea more than that, So in the end I believe that I will do is to say that the dimision of Stab_v^- is at least $\binom{d}{d'+1}$. And by fixing coordinates we add more $\binom{d}{d'}$ restriction. So the dimension becomes:

$$\begin{aligned}&\binom{d}{d'+1} - \binom{d}{d'} \\ &\binom{d}{d-d'+1} - \binom{d}{d-d'} = \binom{d}{d'-1} - \binom{d}{d'}\end{aligned}$$

Note that since we talk on mapping checks $\rightarrow$ bits, we use the ∂^+. Eventually, we need to use the poincare duality $H^{d'} \rightarrow H_{d-d'}$, the duality also should be used to define the SWEEP rule to C_Z. I didn't successes to do it other way.

Claim 1.6. Let v be a vertex. For any $z \in \mathcal{C}_v^{\pm}$, there is a stabilizer $w \in \mathrm{Stab}_v^{\pm}$ such that $z = w|_v$; namely, z and w agree on the faces rooted at v.

Proof. We will show that codes obtained from $\mathrm{Stab}_v^{\pm}$ by forcing the faces rooted at v to $z|_v$ have non-zero dimensions. Then it follows that any code word of $\mathcal{C}_v^{\pm}$ could be extend to codewords in $\mathrm{Stab}_v^{\pm}$. The number of bits in both $\mathrm{Stab}_v^{\pm}$ is:

$$\binom{d}{d'} + d\binom{d-1}{d'}$$

The numbers of checks for $\mathrm{Stab}_{\pm}$ are:

$$\binom{d}{d' \pm 1} + d\binom{d-1}{d' \pm 1}$$

The restriction to $z|_v$ sets the bits rooted at v and satisfies the checks that are rooted at v, so the checks of the form $\theta_{\mathbf{v},\mathbf{S}'}$ for $|S'| = d' \pm 1$ are degenerate, and the dimension that remains is:

$$\dim \geq d\binom{d-1}{d'} - d\binom{d-1}{d' \pm 1}$$

In the case that $d' = (d-1)/2$, then for any $x \in [d-1]$ we have $\binom{d-1}{d'} > \binom{d-1}{x}$ therefore:

$$\begin{aligned} \dim &\geq d\binom{d-1}{d/2} - d\binom{d-1}{d/2 \pm 1} \\ &> d \end{aligned}$$

So there is at least 2^d stabilizers $w \in \mathrm{Stab}_v$ that agree with z. $\square$

Claim 1.7. Let z be a finite assignment. Suppose that

$$\max\left\{L_{d+1}(u) : u \in z\right\} > \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in z\right\} - 1$$

Proof. Let $v_1, v_2, \ldots, v_k$ be the vertices in z that achieve the maximum for L_{d+1}, namely $v_1, v_2, \ldots, v_k = \arg\max_{v \in z} L_{d+1}(v)$, and assume that $v_i \notin \partial^{\pm} z$ for any $i \in [k]$.

Consider the ith vertex v_i. By definition, since $v_i \notin \partial^{\pm} z$, we have $\partial^{\pm} z|_{v_i} = 0$, and since v_i is a maximum point of L_{d+1} in z, it is also a local maximum, and therefore $z|_{v_i+} = z|_{v_i}$. Combining the two, we get that $z|_{v_i}$ is a codeword of the small code at v_i, namely $z|_{v_i} \in \mathcal{C}_{v_i}^{\pm}$. Thus, by Claim 1.6, there is a stabilizer $w_i \in \mathrm{Stab}_{v_i}^{\pm}$ such that w_i and z agree on the positive faces of v_i, namely $w_i|_{v_i} = z|_{v_i}$.

Set $\tilde{z} \leftarrow z + \sum_i w_i$. Since $\sum_i w_i$ is a stabilizer, we have $\tilde{z} \sim_{C_X} z$. In addition, for any $j \neq i$, it follows that $v_j \notin w_i$. Otherwise, there exists $S' \subset S$ such that $v_j \in \theta_{v_i,S'}$, and therefore $L_{d+1}(v_j) > L_{d+1}(v_i)$, in contradiction to v_i and v_j being both maximum points for L_{d+1} in $\partial^{\pm} z$. Thus, we have:

$$\tilde{z}|_{v_i} = (z + \sum_i w_i)|_{v_i} + (z + w_i)|_{v_i} = 0$$

Namely, for any i, the ith vertex v_i is not in $\tilde{z}$. Hence, the vertex domain of $\tilde{z}$ is contained in the union of the vertices of z with the vertices of $w_1, w_2, \ldots, w_k$, substituting $v_1, v_2, \ldots, v_k$. Overall, we get:

$$\begin{aligned}\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} &\leq \max\left\{L_{d+1}(u) : u \in z\bigcup_i w_i / \bigcup_i v_i\right\} \\ &\leq \max\left\{L_{d+1}(u) : u \in z\right\} - 1\end{aligned}$$

□

Claim 1.8. Let z be a finite assignment. Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Proof. Let z be a finite assignment, and let k be an integer such that $k \geq 1$ and

$$\max\left\{L_{d+1}(u) : u \in z\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\} + k$$

For an integer $l \leq k$, consider the assignments $\tilde{z}_0, \tilde{z}_1, .., \tilde{z}_l$ defined as follow:

- The first element, $\tilde{z}_0 = z$.
- If the $i-1$th element satisfies

  $$(\star) \quad \max\left\{L_{d+1}(u) : u \in \tilde{z}_{i-1}\right\} > \max\left\{L_{d+1}(u) : u \in \partial^{\pm} \tilde{z}_{i-1}\right\}$$

  Then define the ith element, $\tilde{z}_i$, to be the result of applying the construction, given in Claim 1.7, on $\tilde{z}_{i-1}$, namely $\partial \tilde{z}_{i+1} = \partial z_i$ and

  $$\max\left\{L_{d+1}(u) : u \in \tilde{z}_i\right\} \leq \max\left\{L_{d+1}(u) : u \in \tilde{z}_{i-1}\right\} - 1$$

  If the $i-1$th element doesn't satisfies $\star$, then set $l = i - 1$.
- If the kth element was defined, set $l = k$.

Since $\sim_{C_X}$ is a transitive relation, we have that for any $i \in [l]$, $z \sim_{C_X} \tilde{z}_i$. If $l \neq k$, then there exists $\tilde{z}_i$ such that $\tilde{z}_i \sim_{C_X}$ and $\tilde{z}_i$ satisfies $\star$, namely by setting $\tilde{z} = \tilde{z}_i$, we get the desired. So, it is left to handle the case where $l = k$. In that case, set $\tilde{z} = \tilde{z}_k$. By Claim 1.7:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Yet, since $\tilde{z} \sim_{C_X} z$, it follows that $\partial \tilde{z} = \partial z$, So:

$$\max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} \tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in z\right\}$$

Therefore, we have the equality:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

□

Claim 1.9. Let z be a finite assignment and let ξ be the result of the application of the Sweep rule on it. Then:

1. For any $i \in [d]$: $\max\left\{L_i(v) : v \in \partial z\right\} = \max\left\{L_i(v) : v \in \partial\xi\right\}$
2. And for $i = d+1$: $\max\left\{L_{d+1}(v) : v \in \partial z\right\} > \max\left\{L_{d+1}(v) : v \in \partial\xi\right\} + 1$

Proof. First, since the Sweep rule is invariant to addition by stabilizers, for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, the correction applied by the Sweep rule on $\tilde{z}$ is the same as the application on z. Denote by ϕ the correction, by $\tilde{\xi}$ the result of applying the Sweep rule on $\tilde{z}$, and by w the stabilizer which is the difference between z and $\tilde{z}$, namely:

$$\begin{aligned}\xi &= z + \phi\\ \tilde{\xi} &= \tilde{z} + \phi\\ \tilde{z} &= z + w\end{aligned}$$

The $\partial^{\pm}$ boundary of $\tilde{\xi}$ equals:

$$\partial\tilde{\xi} = \partial\left(\tilde{z} + \phi\right) = \partial\left(z + \phi\right) = \partial\xi$$

So, in total, we have that for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, it holds that $\partial\tilde{z} = \partial z$ and $\partial\tilde{\xi} = \partial\xi$. Therefore, showing that there exists $\tilde{z} \sim_{C_X} z$ for which the claim holds proves the claim for z. By Claim 1.8, there is $\tilde{z} \sim_{C_X} z$ such that:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm}\tilde{z}\right\}$$

For such $\tilde{z}$, any maximum of L_{d+1} in $\partial\tilde{z}$, is also a maximum in $\tilde{z}$. Denote by $v_1, .., v_k$ the maximum points in $\partial\tilde{z}$. Since they are maximum points in $\tilde{z}$, it holds that for each $i \in [k]$, $\tilde{z}|_{v_i} = \tilde{z}|_{v_i+}$, and hence $\partial\tilde{z}|_{v_i} = \partial\tilde{z}|_{v_i+}$. Since $\tilde{z}|_{v_i+}$ is an assignment rooted at v_i, then v_i can suggest $\tilde{z}|_{v_i+}$, namely the condtion in Sweep rule is satisfied for vertex v_i. Thus:

$$\partial\tilde{\xi}|_{v_i} = \partial\tilde{z}|_{v_i} + \partial\phi|_{v_i} = 0$$

Namely, for any i, the ith vertex v_i is not in $\tilde{\xi}$. For any vertex v denote by ϕ_v the correction made by the vertex v, namely $\phi = \sum_{v\in\partial\tilde{z}} \phi_v$. Hence, the vertex domain of $\tilde{\xi}$ is contained in the union of the vertices of $\tilde{z}$ with the vertices of $\cup_{v\in V}\phi_v$, substituting maximum points $v_1, v_2, \ldots, v_k$. Overall, we get:

$$\begin{aligned}\max\left\{L_{d+1}(u) : u \in \partial\tilde{\xi}\right\} &\leq \max\left\{L_{d+1}(u) : u \in \partial\tilde{z} \bigcup_{v\in\partial\tilde{z}} \phi_v / \bigcup_i v_i\right\}\\ &\leq \max\left\{L_{d+1}(u) : u \in \partial\tilde{z}\right\} - 1\end{aligned}$$

Let v be a vertex that is a maximum of L_g in $\partial\tilde{z}$, maximum points of L_g do not see syndrome at the g direction. If and:

$$\phi_v = \sum \theta_{\mathbf{v},\mathbf{S}'_\mathbf{i}}$$

By Claim 1.3 $\partial \tilde{z}|_{gv} = 0$, Hence:

$$\begin{aligned}
\partial \phi_v|_{gv} &= \left(\partial \sum \theta_{\mathbf{v},\mathbf{S'_i}}\right)|_{gv} \\
&= \left(\sum_{S'_i} \sum_{g' \in S'_i} \theta_{\mathbf{v},\mathbf{S'_i}/\mathbf{g'}} + \theta_{\mathbf{g'v},\mathbf{S'_i}/\mathbf{g'}}\right)|_{gv} \\
&= 0
\end{aligned}$$

Thus if $g \in \cup S'_i$, Then:

$$\theta_{\mathbf{gv},\mathbf{S'_i}/\mathbf{g}} \in \partial \phi_v|_{gv}$$

So for $\theta_{\mathbf{gv},\mathbf{S'_i}/\mathbf{g}} = 0$, it must appear at least twice in the sum, yet the left terms in the sum cannot contain it (they differ by the root vertex), and if there is a right term $\theta_{\mathbf{gv},\mathbf{S'_j}/\mathbf{g}} = \theta_{\mathbf{gv},\mathbf{S'_i}/\mathbf{g}}$, then it implies that $S_j = S_i$. Which is a contradiction, therefore: $g \notin \cup S'_i$ and $gv \notin \phi_v$.

$$\begin{aligned}
\max\left\{L_g(u) : u \in \partial \tilde{\xi}\right\} &\leq \max\left\{L_g(u) : u \in \partial \tilde{z} \bigcup_{v \in \partial \tilde{z}} \phi_v / \bigcup_i v_i\right\} \\
&\leq \max\left\{L_g(u) : u \in \partial \tilde{z}\right\} + 1 - 1
\end{aligned}$$

$\square$